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Medical Image Character Recognition Based on Multi-scale Neural Convolutional Network

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***Abstract*—Generally, medical images are usually characterized by complex background and low resolution. Also, text's location is sometimes random. Therefore, traditional Optical Character Recognition (OCR) technology cannot be well applied to recognize medical images which contain many factors. Aiming at this problem, a character recognition algorithm for medical images is proposed in this paper. The Efficient and Accurate Scene Text (EAST) scene detection algorithm is used in text detection and the Convolutional Recurrent Neural Network (CRNN) model is used in character recognition. The problem of low recognition rate that caused by the low resolution will be solved by adding a multi-scale convolution kernel to the model. The designed algorithm is tested by Medpix datasets, and the experimental results show that the proposed method has a higher precision and a higher recall than the traditional recognition methods.**

Keywords—medical image, character recognition, text detection, revolution recurrent neural network

I. INTRODUCTION

In the field of medical imaging, with the rapid development of artificial intelligence technology, image recognition and processing technology is more and more widely used for medical image processing. Character recognition and translation have become the basic needs of daily medical treatment. However, most images have heavy background interference and low image definition, the accuracy of traditional recognition methods will be affected when retrieving and using medical images.

At present, the commonly used character recognition algorithms include Tesseract Traditional optical character recognition (OCR), Dense Convolutional Network (DenseNet OCR), Convolutional Recurrent Neural Network (CRNN), etc. Tesseract OCR algorithm is designed based on the function of TessBaseAPI and consists of two procedures.[1] The first step is to divide the image and determine the text area. The second step is to cut out the characters individually, and then recognize them one by one. However, as one of the earliest character recognition algorithm, it runs slowly and has low recognition accuracy. As an end-to-end recognition method, CRNN[2], which is composed of Convolutional Neural Network (CNN), Recurrent Neural Networks (RNN) and Connectionist Temporal Classification (CTC) modules, can be well adapted to different length of text. In the general natural scene, it can satisfy the requirements of high precision and high speed. Efficient and Accurate Scene Text (EAST)[3] model is a fast and accurate text detection model, which is composed of full convolution network model and non-maximum suppression algorithm. Advanced EAST model is an improvement of EAST to recognize longer text. DenseNet OCR[4]connects the features to form a linked list when extracting eigenvalues. Therefore, it is more conducive to the transmission of information because of the fewer parameters.

For conventional medical images, there are four common problems. The first common problem is that the text characters are too small, resulting in low resolution when selecting text regions for processing, as shown in Fig. 1.

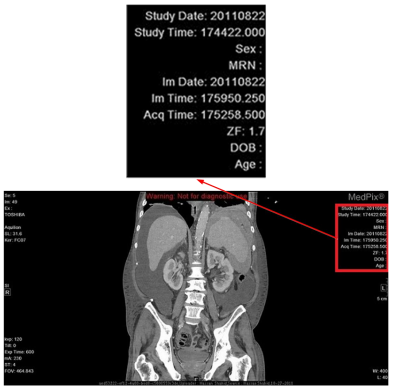


Fig. 1. Texts in medical images.



Fig. 2. Background interference for the medical image character recognition.

When the marked part is enlarged, the resolution of the text is reduced. This kind of fuzzy text is difficult to recognize in traditional methods.

The second common problem is randomness of text position. It causes a certain deviation in the positioning during the character recognition process, thereby affecting the accuracy of character recognition.

The third common problem is the influence of special characters. In the character recognition process, the special characters are mistaken for the similar characters, such character recognition performance is deteriorated.

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The last common problem is the background interference. Interference information will make the identification process difficult, as shown in Fig. 2. The text marked in the picture is complex in background, and similar in color to the image.

Owing to these problems, the traditional character recognition scheme cannot be fully applied to medical images directly. This paper presents a new character recognition algorithm based on multi-scale convolutional neural network.

The problem of background interference is solved by data enhancement, and multi-scale convolution kernels are added to solve the problem of low resolution. That is to say, the character recognition in medical image is realized by constructing the models of EAST and CRNN. Experimental results show that the proposed method has significant advantages in medical image character recognition.

II. Basic Framework of Character Recognition

The framework of the proposed method is based on EAST scene text detection algorithm and CRNN text recognition algorithm. The recognition procedure consists of two steps. First, text detection algorithm locates the text to be recognized, and then CRNN model recognizes the located text and outputs the results. The specific framework is shown in Fig. 5.

III. Text Location Box Selection and EAST Model

EAST is an efficient and accurate scene text detection algorithm, which contains two modules, in which the first one is a full convolution network model, which forecasts the object text and selects the text to be recognized. The second module uses a non-maximum suppression algorithm to filter the existing text box and output the final results.

A. Full convolution network of EAST model

The full convolution network consists of three parts: feature extraction, feature fusion and output layer. Due to the small receptive field, the traditional EAST model is easy to fail in detecting long text. However, there are some long texts in medical images. In order to avoid the problems of recognition error or missing information in the detection of these long texts, the traditional EAST model needs to be modified to adapt to the detection of long text.

In this paper, the output layer of the model is improved by predicting head and tail elements to realize the detection of long texts. The specific framework is shown in Fig. 3.

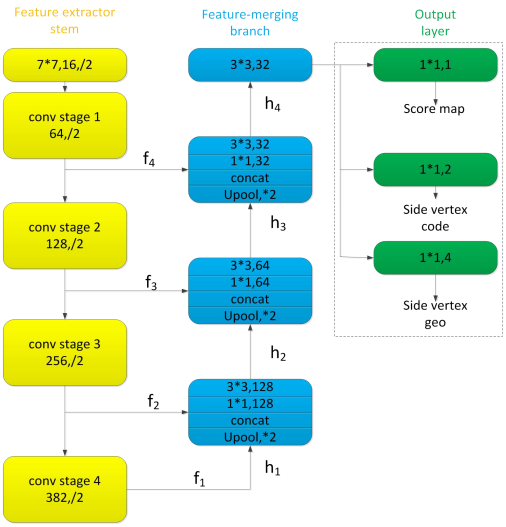


Fig. 3. Framework of full convolution network of EAST model with changed output layer.

Feature extraction: Using the existing initialization of training parameters, feature images with sizes of 1/32, 1/16, 1/8 and 1/4 are extracted from the image to be processed.

Feature merging: The merging mode is layer by layer. During each merging stage, the feature map from the previous stage is first input into an unpooling layer, and expanded to twice the original size. Then, it merges with the current layer feature map (channel). Finally, a 1*1 convolution is used to reduce the number of channels and the amount of computation, and the local information is fused to produce the final output of the merging stage.

Output layer: The final output information includes: the confidence level of the detection box, “score map”, confidence of boundary elements, head and tail elements, “side vertex code”, the four vertices predicted by the head and tail elements, “side vertex geo”. This method use the adjacent activation point to determine the region list, and then the adjacent region lists form the region group. With this process, the prediction of text rotation angle in the traditional EAST model is omitted. And the amount of calculation will also be reduced.

In this model, the head (tail) element is closer to the two vertices of the head (tail) boundary. When using the head (tail) element to predict the head (tail) vertex, there is no prediction failure due to the distance beyond the receiving range. Normally, the words in medical images are from left to right. This paper directly sets the left side of the text as the head element and the right side as the tail element, which can reduce the amount of calculation. Through this improvement, long text can be accurately detected, and the effect is shown in Fig. 4.

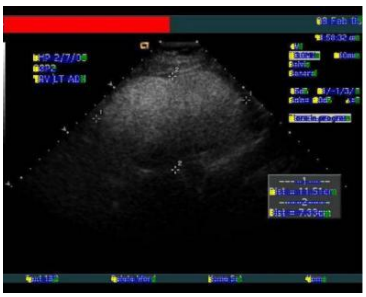


Fig. 4. Text detection results.

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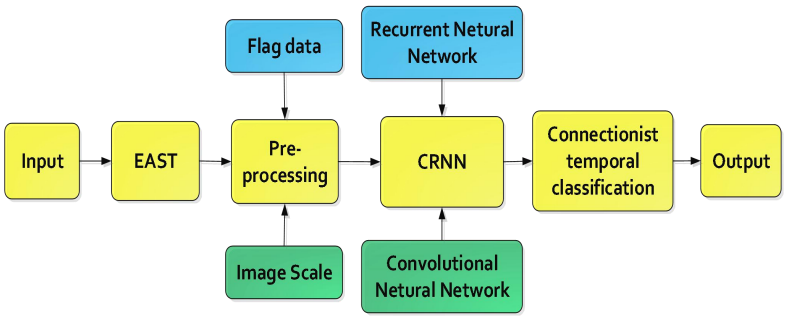


Fig. 5. Framework of character recognition.

B. *Non-maximum suppression algorithm in EAST*

Non maximum suppression algorithm (NMS) is a common post-processing algorithm for target detection. Its main function is to filter the text detection box, reject the detection box with poor accuracy, and select the optimal detection box. At present, the common NMS algorithms mainly include: traditional NMS[5], IoU guided NMS[6], Adaptive NMS[7], etc. In this paper, the soft NMS[8] algorithm is selected based on the classification priority. The procedure of the algorithm is as follows.

First, a set S is selected to store the text box to be processed. Then, all the test frames are handed in and ranked according to the confidence scores, and the detection box M with the highest score is selected. The score of M will be taken as the input of the function. Then the IoU (Intersection over Union) of M and the remaining detection boxes will be calculated , and the detection boxes that exceed the threshold will be deleted. This process will be applied recursively to other diagrams. The process is as Fig.6.

In NMS algorithm M is the current maximum score box, it will be set to zero directly if the corresponding value of intersection over union (IoU) exceeds the threshold. However, the screening mechanism is too strict that will lead to unfriendliness to some cases with non-simple background. The Soft NMS algorithm selected in this paper is a smoother method, it improves the original elimination algorithm to linear penalty.

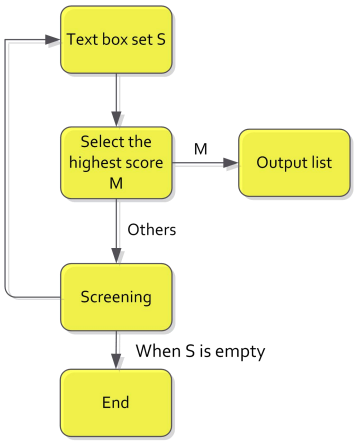


Fig. 6. NMS algorithm flow.

Soft NMS algorithm has the same complexity as traditional NMS algorithm, and it is more flexible and easier to implement. It is more efficient and convenient, and can be used in most detection processes.

IV. Improvement of CRNN Recognition Model

Aiming at the dilemma of character recognition in current medical scene, an improved CRNN network model is proposed to solve the problems in the application of traditional recognition methods. The main ideas are as follows.

First, aiming at the problem of complex text background and the interference factors in image region, the method of data enhancement is selected, and the targeted model training is carried out. There are many ways to enhance data, such as zoom, rotate, flip, blur, image blending, etc. Considering the complexity of medical image background, image blending is chosen as the main method. A picture with a complex image and another picture with text were chosen and given different weights to make them transparent, then mixed together to make a new picture, as shown in Fig. 7.

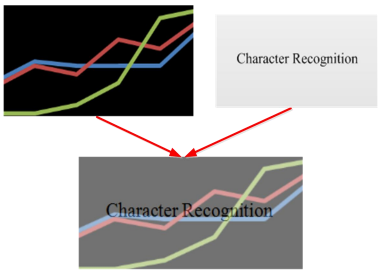


Fig. 7. Image blending Data Augmentation.

As can be seen from Fig. 7, the upper left is an image composed of three curves of different colors on a black background from the bottom left to the upper right path, and the upper right is an image composed of the two characters “Character Recognition” on a gray background. Combining these two images with different weights to form a new image, the new image contains three curves, two characters, and gray and black backgrounds in the original image. Such an imagefusion method can enhance the data to deal with complex text backgrounds and interference factors.

Based on the data enhancement method, data sets are built. This paper attempts to simulate the complex background similar to medical images. This model has a significant improvement in the processing of background interference, through this kind of image training. The chart of model training is shown in Fig.

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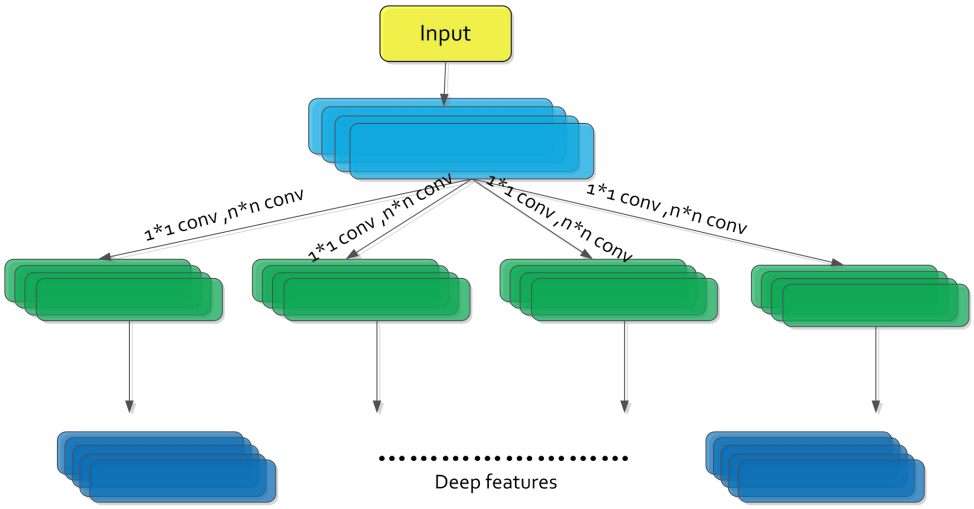


Fig. 8. Feature extraction

As can be seen from Fig. 9, by using the data set to perform data enhancement and image blending, this step completed the construction of the image data set under the complex background. Based on the model training process, the image data set is analyzed and recognized, the feature extraction of the image data set is realized, and the character is completed for identifying the task. Furthermore, several convolution kernels with different scales are introduced to solve the problem of low resolution of template image caused by small image texts.

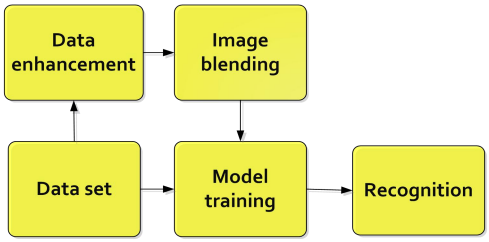


Fig. 9. Flow chart of model training.

In feature extraction, multiscale convolution is introduced to ensure abundant image information by fusing convolution kernels of different fields. And then more complete image features are extracted to realize the accurate recognition of the recognition target. The feature extraction is shown in Fig.8.

When the picture is input to the network, the convolutional layers of different structures are used to extract the features of it. Such a structure increases the width of the network, thereby obtaining image features of different dimensions. However, it will also increase the number of network parameters and affect the speed of training and testing. Therefore, in order to balance the contradiction between the two steps, after the image is input to the network, the 1*1 convolution is used to adjust the number of channels, and then the convolutional layer is used to extract features.

V. Experimental Results

The configuration of the experimental environment is as follows: Intel Core i7 processor, 64G memory, GTX 1080 Ti graphics card, the design system of Ubuntu 16.04.2, and the development language of Python 3.8.3.

The application results of the improved character recognition algorithm in medical images are shown in the figure below.

As can be seen from Fig. 10, the constructed model has a good effect in dealing with low resolution text. It is proved that this model can be used to recognize fuzzy words in medical images.

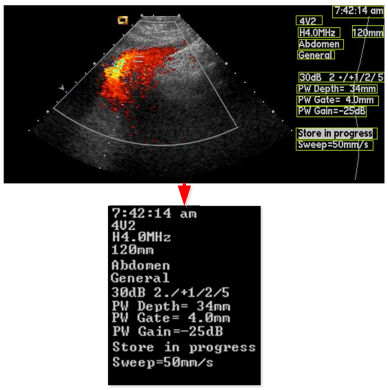


Fig. 10. Recognition results of low resolution image

The experimental results in Fig. 11 show that the constructed model can deal with the interference of complex background in the image.

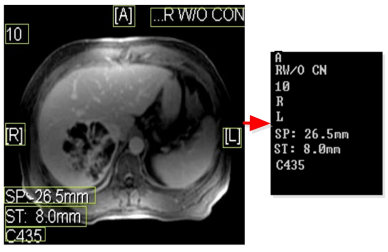


Fig. 11. Recognition results of complex background image

The medical image recognition method based on our improved CRNN network model is compared with CRNN method on MedpixCardiac Atlas (cardiac MRI image) experimental data set and Mjsynth data set respectively.

Mjsynth data set is a commonly used composite dataset, which contains 8.9 million sample images, among then 1500

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pictures were selected for training and 1000 pictures for testing.

MedpixCardiac Atlas (cardiac MRI image) experimental data set is an open source medical image data set, 1500 images were selected for training and 1000 images for testing. The results are shown in TABLES I and Π.

TABLE I. COMPARITION RESULTS BY MJSYNTH DATA SET

Method	Precision	Recall	F1-measure
CRNN	0.78	0.73	0.75
OURS	0.80	0.73	0.76

TABLE II. COMPARITION RESULTS BY MEDPIX DATA SET

Method	Precision	Recall	F1-measure
CRNN	0.64	0.61	0.62
OURS	0.70	0.65	0.67

Compared with the CRNN network model which performed well in Mjsynth data set, the Precision, Recall rate and F1-measure are close to each other. However, in Medpix data set, there are obvious differences with CRNN. The improvement of Precision, Recall and F1-measure is 6%, 4% and 5% respectively. The results show that the modified recognition algorithm is almost the same as the traditional recognition one in the common natural background. However, it has an obvious advantage in recognition accuracy and efficiency, compared with several traditional methods in the medical image background.

VI. CONCLUSION

In this paper, a character recognition method based on medical image background has been proposed, which can effectively solve the problems of low resolution and background interference in the process of medical image processing. Through multi-scale extraction and fusion of feature image, the information of the detected image is obtained more effectively, and the convolution neural network is optimized for the prediction and extraction of the image text. Compared with the commonly used character recognition schemes, the experimental results show that in the conventional background, the improved character recognition algorithm is not much different from the traditional ones. However, the improved recognition algorithm has significant advantages over other algorithms in the medical image background.

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